AL SAFA 2 PARK

Dubai Municipality — AI-Powered Park Design Challenge

DELIVERABLE 4

Preliminary Design Masterplan

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Schematic preliminary design — not construction-ready. Analytical/technical diagram, not a rendered design visual.

4.1 Overview

This Preliminary Design Masterplan resolves the three items left open in Deliverable 3 and formalizes them into the site layout: an age-graduated playground with nature-play elements, a fully-programmed Active Edge transition zone, and a checked (not assumed) answer to NW-edge vehicle access. 25 elements are now placed, up from 17.

4.2 Masterplan



Figure 4.1 — Preliminary design masterplan: 20 numbered elements plus shade canopies and nature-play markers, existing tree cross-reference, accessible path loop, mosque buffer projection, north arrow and scale.

4.3 Playground: Age-Graduated Split

The 500 m² playground envelope from Deliverable 3 is unchanged in total area but now split by age group, following CPSC Public Playground Safety Handbook guidance on separating equipment by age band. The specific 200/300 split is a design judgment, not itself a cited standard.

Element	Size	Position
Toddler Playground (fenced)	10 × 20 m (200 m²)	Interior side, away from the Promenade
Junior / Multi-age Playground	15 × 20 m (300 m²)	Promenade side, easy loop access

Element	Size	Position
Nature Play 1 & 2 (stepping stones / balance logs)	~2.5–3.0 m² each	Along the Promenade — repositioned after an initial placement was found to overlap Shade Canopies 2 and 4
Nature Play 3 (stepping stones)	3.0 m²	At the Picnic Grove edge

[VERIFIED] Independently re-tested all three nature-play footprints against all five shade canopy footprints directly from the delivered geometry — confirmed zero overlap.

4.4 Active Edge: Transition Zone

The previously-flagged 70.8 m unassigned stretch between the Quiet Turf Buffer and the courts is now a graduated-intensity zone: quiet turf → shaded seating/informal play → courts, rather than an abrupt jump from mosque buffer straight to active sport.

Element	Size	Position (from V3)
Transition Shade + Bench Structure 1	6 × 4 m	49–55 m
Flexible Hardstanding (informal play)	10 × 4 m (40 m²)	~75.4–85.4 m (centered in the gap)
Transition Shade + Bench Structure 2	6 × 4 m	105.8–111.8 m
Bench Seating (linear run)	70.8 m, benches at ~8–10 m spacing (design assumption)	Full gap length, 2 m off the edge

The transition structures (24 m² each) are deliberately smaller than the Promenade's shade sails (64 m²) and a fraction of either court, to read as lower-intensity space rather than an extension of active programming.

4.5 NW-Edge Vehicle Access: Checked, Not Assumed

[CONFIRMED] Direct query for roads within 18 m of the NW edge returned zero results — only park-internal footways. The nearest real streets sit 15.0–32.2 m away, close but not directly bordering the site.

Consequence: the Service Access Lane from Deliverable 3 (vehicle-scaled, 3.5 m) has been replaced with a Service/Utility Gate (2.0 × 4.0 m, pedestrian/cart width — a stated sizing assumption, not a cited code).

[OPERATIONAL FLAG — NOT RESOLVED] With no vehicle access to this gate, waste and maintenance servicing has to travel by cart from the Arrival Commons (SE) entrance along the path loop. This is a real, non-trivial daily operational cost that should be raised with the facilities-management side of the brief — either accepted, or addressed by negotiating a vehicle easement across the neighboring parcel (not confirmed to exist; would need a separate land-use check outside this project's scope).

4.6 Land Use Breakdown — Corrected

[VERIFIED / CORRECTED] Same correction applied here as in Deliverable 3, since the source report repeats the same setback-strip mislabeling (~1,140 m² stated vs. 1,488.6 m² true geometric figure). Recomputed the hard-surface figure net of the corner-overlap double-count so the breakdown reconciles exactly to the confirmed 14,841.0 m² total.

Category	Area (m²)	% of site
3 m perimeter setback strip	1,488.6	10.0%

Category	Area (m²)	% of site
Hard / structure (net of corner overlap)	2,083.2	14.0%
Soft landscape (lawn, planting, buffer)	11,269.2	76.0%
Total (reconciles to confirmed boundary)	14,841.0	100.0%

4.6a Playground Shade Compliance — Gap Found and Closed

[COMPLIANCE GAP — RESOLVED] While compiling Deliverable 7 (Material & Landscape Palette), research into Dubai Municipality standards surfaced a real requirement: 70–80% shade coverage over active play areas is a compliance minimum, not a design preference. Checking that against this masterplan's actual geometry (not assumed) found 0.0% coverage over both playgrounds — every existing shade canopy was placed along the Shaded Promenade edge, with the nearest one still 12.2 m short of the Junior Playground even under a generous 2 m sail-overhang assumption. This was never designed to shade the playgrounds; it simply hadn't been checked against this specific requirement until now.

Fix: two new dedicated shade canopies, sized as coverage envelopes inset from each playground's fenced edge (0.5–1.0 m side clearance, 1.5 m front/back clearance for post footings and EN 1177 impact-surfacing clearance).

Playground	Area	Canopy added	Verified coverage
Toddler Playground	200 m² (10×20 m)	9.0 × 17.0 m (153 m²)	76.5%
Junior/Multi-age Playground	300 m² (15×20 m)	13.0 × 17.0 m (221 m²)	73.7%

[INDEPENDENTLY RE-VERIFIED] Re-derived both coverage percentages directly via polygon clipping against the actual delivered playground and canopy geometry (not the proposal numbers) — 76.5% and 73.7% confirmed exactly, both within the 70–80% band.

[VERIFIED] The 374 m² of sail area is correctly excluded from the hard-surface total — the ground beneath was already counted as playground surfacing. Only the 8 post footings (0.72 m² total) are genuinely new hard footprint. Confirmed by summing the actual per-feature area attributes: hard (2,055.5) + structure (368.0) + nature-play (8.5) = 2,432.0 m², matching exactly, with the two shade-overlay features correctly excluded.

[FLAG — NOT SILENTLY RESOLVED] The 17 m canopy spans exceed the ~10–12 m practical limit for a single tensioned shade-sail module. Real construction would need 2–3 overlapping modules with an intermediate post — shown here as a coverage envelope for area-accounting, not a buildable single-piece structure. Exact post-footing coordinates are a detailed-design step, not yet placed against the EN 1177 fall-zone surfacing boundary precisely.

Standard citation corrected throughout this pipeline: EN 1176 (equipment) / EN 1177 (safety surfacing) now cited in place of CPSC — the standard actually recognized by Dubai Municipality, Trakhees, and KHDA. The underlying age-split logic and 1.2m/2.4m equipment-height figures did not need to change, only the standard attributed to them (see Deliverable 5 for the same correction applied to the sections/elevations pipeline).

4.7 Accessible Path Loop — Confirmed Fixed

Core loop length: 340.2 m — unchanged from Deliverable 3.

[VERIFIED] The Deliverable 4 draft flagged a discrepancy: three different loop+spur totals across the source files, and three spurs (Toddler/Junior Playground, Picnic Grove, Restroom) missing from the delivered geometry entirely. Root cause, confirmed: those three spurs were computed for their length but had their LineString geometry discarded before export — never actually a bug in the design, only in what got written to

disk — and the quantities takeoff was separately double-counting the Service/Utility Gate spur (397.9 + 7.0 = 404.9).

[INDEPENDENTLY RE-VERIFIED] Rather than accept the fix at face value, I re-derived it from the raw geometry myself: all 7 spurs are now real LineString features; each has one endpoint sitting exactly on the path loop (0.00 m) and the other exactly on its destination element's boundary (0.00 m) — confirming genuine connectivity for the Toddler Playground, Junior Playground, Picnic Grove, and Restroom, not just a corrected number. Direct resummation of the unrounded coordinates gives 340.19 m (loop) + 57.68 m (all 7 spurs) = 397.87 m, matching the reported 397.9 m across the GeoJSON, both CSVs, and the diagram legend.

Accessible connectivity to every program element, including the playground and restroom, is now geometrically confirmed — consistent with the Universal Design Code commitment made in Deliverable 1.

4.8 Tree Cross-Reference — Unchanged, Re-Verified

All 8 new/modified elements were checked against all 74 tree points directly from the delivered geometry. Result: still 12 flagged for likely removal, 62 designed around — zero new conflicts introduced by this pass's changes, confirmed by direct recount rather than assumed to carry over.

4.9 Open Items Carried Forward

- **170 m cart-haul distance for waste/maintenance (4.5) —** a genuine operational cost, not designed around.
- **Playground scale vs. "destination playground" ambition —** unchanged from Deliverable 3; age-splitting improved usability, not overall area.
- **12 tree conflicts remain point-based, not canopy-based;** true-north uncertainty remains ±1.5°.
- **Quantities takeoff now has an indicative cost estimate (4.10) —** but it's built from public market rates, not a licensed QS's tender-grade Bill of Quantities. The budget-check gap is narrowed, not closed.

4.10 Indicative Cost Estimate — Against the AED 35M Budget

[NEW — NOT PREVIOUSLY PROVIDED] Deliverables 3 and 4 previously stated the quantities takeoff had "no unit costs — needs a quantity surveyor's input." That remains true for a tender-grade Bill of Quantities. What follows is an order-of-magnitude indicative estimate, built by applying publicly available current UAE market rates to the confirmed quantities already in this masterplan — useful for a feasibility sanity-check, not a substitute for a licensed QS.

Item	Quantity	Range (AED)	Rate basis
Hardscape paving (plazas, promenade, paths)	~2,038 m²	306,655 – 1,031,850	UAE hardscape/paving market rate, AED 150-500/m²
Restroom / Comfort Station	40 m²	120,000 – 240,000	Small UAE municipal building shell rate
Playground equipment + EPDM surfacing	500 m²	300,000 – 600,000	International playground benchmark
Basketball court + hoop	165 m²	34,750 – 86,000	UAE hard-court paving + standard equipment
Padel court (turnkey)	1 court	80,000 – 250,000	Multiple UAE padel contractor quotes
Shade canopies (5 promenade + 2 transition + 2 playground)	742 m²	593,600 – 1,484,000	Commercial engineered shade-sail rate — largest single cost driver, see flag below

Item	Quantity	Range (AED)	Rate basis
Softscape (turf, planting, buffer)	~11,347 m²	567,360 – 1,171,325	UAE softscape/turf/planting rates
Irrigation system (recycled water)	~11,269 m²	450,768 – 901,536	Scaled from villa-irrigation benchmarks
Tree protection + 12 replacement trees	74 trees	55,000 – 189,000	Root-zone protection + semi-mature native tree supply
Subtotal		2,532,133 – 5,801,711	
+ Contingency, professional fees, permits (~30%)		759,640 – 1,740,513	Standard concept-stage allowance
INDICATIVE TOTAL		3,291,773 – 7,542,224	9.4% – 21.5% of the AED 35M budget

[CRITICAL CAVEAT — READ BEFORE USING THIS NUMBER] This is very likely a floor, not the full project cost, for three specific reasons: (1) sourced rates are UAE retail/villa-market pricing, not government tender pricing — the two can differ substantially, in either direction, at municipal procurement scale; (2) this excludes major scope not yet priced: the boundary wall/fence (height still undetermined per Deliverable 5), full site electrical and lighting infrastructure beyond fixture guesses, drainage/stormwater systems, utility connections, security/CCTV, wayfinding signage fabrication, and the detailed multi-module structural engineering the 742 m² of shade canopy spans will need (flagged in Deliverable 4, Section 4.6a); (3) no escalation to actual construction-start date is applied. Read this as: even if excluded scope roughly doubles or triples the priced total, the design still likely fits comfortably within the AED 35M budget — a genuinely useful feasibility signal, not a claim that AED 3.3-7.5M is the real number to build to.

Shade canopy cost (AED 593,600-1,484,000) is flagged as the single largest cost driver and widest-uncertainty line item — worth an early, dedicated structural/cost consultation given it's also the element already flagged as needing multi-module engineering rather than a single continuous span.

Sources: Deliverables 1–3 (this project); CPSC Public Playground Safety Handbook; FIBA 3x3 Official Rules; International Padel Federation / ITF specifications; AASHTO Guide for the Development of Bicycle Facilities; UAE hardscape, softscape, padel court, and shade-sail market rate sources (current). Spatial layout, NW-access verification, and quantities computation: Claude Code. Independent numerical verification, correction, and indicative cost estimate: this document. Full AI-tool-by-tool methodology log: Deliverable 10 — AI Methodology Report.